



Single Node Optimisation Vectorisation



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Vector Instructions (Vectorisation)

- Modern CPUs can perform multiple operations each cycle
 - Use special SIMD (Single Instruction Multiple Data) instructions
 - e.g. SSE, AVX
 - Operate on a "vector" of data
 - typically 2 or 4 double precision floats (on AMD Rome)
 - But can be up to 8 per FPU
 - Potentially gives speedup in floating point operations
 - Usually only one loop is vectorisable in loop nest
 - And most compilers only consider inner loop

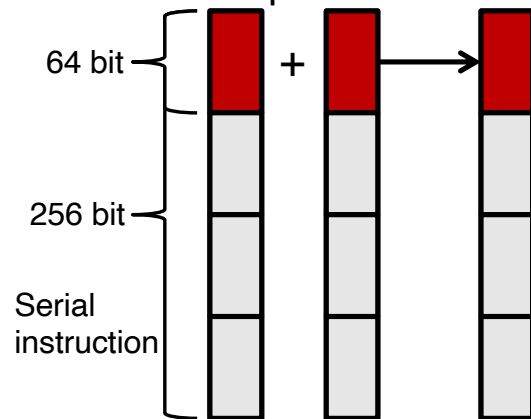
Vectorisation

- Same operation on multiple data items

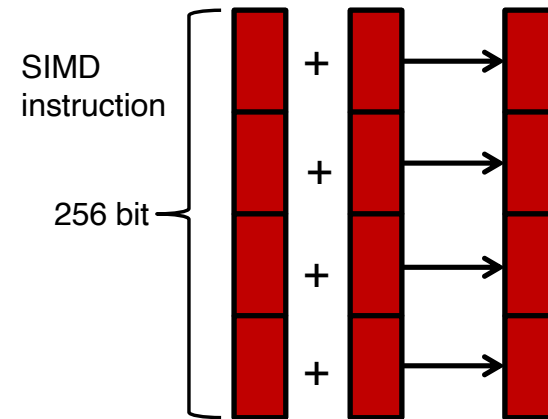
- Wide registers
- SIMD needed to approach FLOP peak performance, but your code must be capable of vectorisation

- **x86 SIMD instruction sets:**

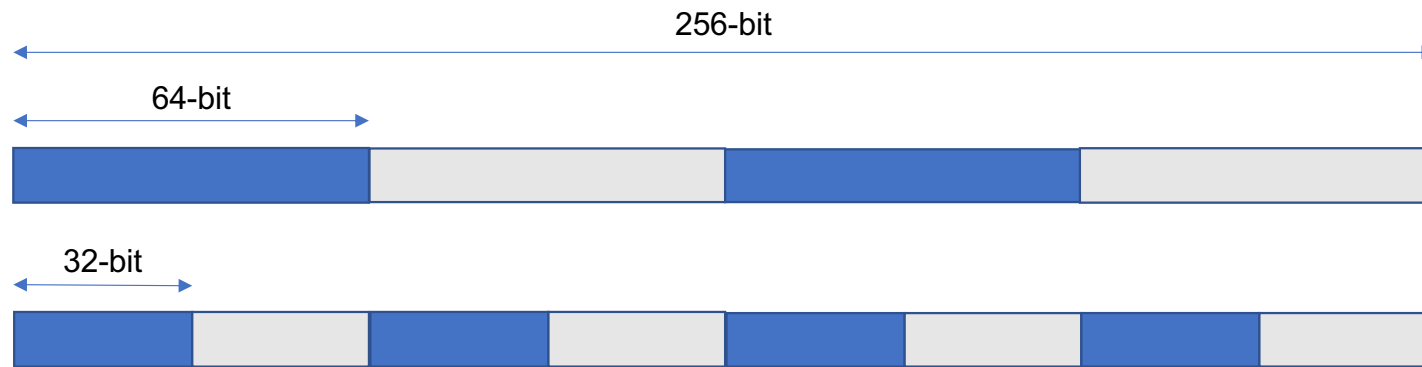
- SSE: register width = 128 Bit
 - 2 double precision floating point operands
- AVX: register width = 256 Bit
 - 4 double precision floating point operands



```
for(i=0;i<N;i++){  
    a[i] = b[i] + c[i];  
}  
do i=1,N  
    a(i) = b(i) + c(i)  
end do
```



AVX2/AVX256



- Rome processor has two AVX256 vector units per core
 - Symmetrical units
 - Only one supports some of the legacy stuff (x87, MMX, some of the SSE stuff)
 - Vector instructions have a latency of 6 cycles

- Optimising compilers will use vector instructions
 - Relies on code being vectorisable
 - ...or in a form that the compiler can convert to be vectorisable
 - Some compilers are better at this than others
 - But there are some general guidelines about what is likely to work...

When does the compiler vectorize

- What can be vectorized
 - Only loops
 - Usually only one loop is vectorisable in loopnest
 - And most compilers only consider inner loop
 - Optimising compilers will use vector instructions
 - Relies on code being vectorisable
 - Or in a form that the compiler can convert to be vectorisable
 - Some compilers are better at this than others
 - Check the compiler output listing and/or assembler listing
 - Look for packed AVX/AVX2/AVX512 instructions
- i.e. Instructions using registers `zmm0–zmm31` (512-bit) `ymm0–ymm31` (256-bit) `xmm0–xmm31` (128-bit)
- Instructions like `vaddps`, `vmulps`, etc...

Requirements for vectorisation



- Loops must have determinable (at run time) trip count
 - rules out most while loops
- Loops must not contain function/subroutine calls
 - unless the call can be inlined by the compiler
 - maths library functions usually OK (but not with GNU compilers)
- Loops must not contain branches or jumps
 - guarded assignments may be OK
 - e.g. `if (a[i] != 0.0) b[i] = c * a[i];`
- Loop trip counts needs to be long, or else a multiple of the vector length



- Loops must not have dependencies between iterations
 - reductions usually OK, e.g. `sum += a[i];`
 - avoid induction variables e.g. `indx += 3;`
 - use **restrict**
 - may need to tell the compiler if it can't work it out for itself
- Aligned data is best
 - e.g. AVX vector loads/stores operate most effectively on 32-bytes aligned address
 - need to either let the compiler align the data....
 - ..or tell it what the alignment is
- Unit stride through memory is best

Compilers

- Cray (C) and AMD compilers requires
 - Optimisation enabled (generally is by default)
 - `-O2`
 - To know what hardware it's compiling for
 - `-march=znver2`
 - This is added automatically for you on ARCHER2
 - Can disable vectorisation
 - `-fno-vectorize`
 - Useful for checking performance
 - Cray compiler will provide vectorisation information
 - `-Rpass-missed=loop-vectorize -Rpass-analysis=loop-vectorize`
 - Other compilers information
 - Cray Fortran: `-hlist=a`
 - GNU: `-fdump-tree-vect-all=<filename>`



Did my loop get vectorised?

- Always check the compiler output to see what it did
 - CCE: `-hlist=a`
 - GNU: `-fdump-tree-vect-all=<filename>`
 - AMD: `-Rpass-missed=loop-vectorize -Rpass-analysis=loop-vectorize`
 - or (for the hard core) check the assembler generated
 - Look to see which registers are in use.
- Clues from CrayPAT's HWPC measurements
 - `export PAT_RT_HWPC=13` or `14` # Floating point operations SP,DP
 - Complicated, but look for ratio of operations/instructions > 1
 - expect 4 for pure AVX with double precision floats

Did my loop get vectorised?

- GNU offers other options for checking:
- `-fopt-info`
- `-fopt-info-all`
- `-O3 -fopt-info-missed=missed.all`
- `-O2 -ftree-vectorize -fopt-info-vec-missed`
- `-fopt-info-loop-optimized`

Helping vectorisation

- Does the loop have dependencies?
 - information carried between iterations
 - e.g. `counter: total = total + a(i)`
 - No:
 - Tell the compiler that it is safe to vectorise
 - Yes:
 - Rewrite code to use algorithm without dependencies, e.g.
 - promote loop scalars to vectors (single dimension array)
 - use calculated values (based on loop index) rather than iterated counters, e.g.
 - Replace: `count = count + 2; a(count) = ...`
 - By: `a(2*i) = ...`
 - move `if` statements outside the inner loop
 - may need temporary vectors to do this (otherwise use masking operations)
- Is there a good reason for this?
 - There is an overhead in setting up vectorisation; maybe it's not worth it
 - Could you unroll inner (or outer) loop to provide more work?

Vectorisation example

- Compiler cannot easily vectorise:
 - Loops with pointers
 - Non-unit stride loops
 - Funny memory patterns
 - Unaligned data accesses
 - Conditionals/Function calls in loops
 - Data dependencies between loop iterations
 -

```
int *loop_size;
void problem_function(float *data1, float *data2, float *data3, int *index){
    int i,j;
    for(i=0;i<*loop_size;i++){
        j = index[i];
        data1[j] = data2[i] * data3[i];
    }
}
```



Vectorisation example

- Can help compiler
 - Tell it loops are independent
 - `#pragma clang loop vectorize(enable)`
 - `-Menable-vectorize-pragmas !dir$ ivdep`
 - Tell it that variables or arrays are unique
 - `restrict`
 - Align arrays to cache line boundaries
 - Tell the compiler the arrays are aligned
 - Make loop sizes explicit to the compiler
 - Ensure loops are big enough to vectorise

```
int *loop_size;

void problem_function(float * restrict data1, float * restrict data2, float * restrict data3, int
* restrict index){
    int i,j,n;
    n = *loop_size;
    #pragma ivdep
    for(i=0;i<n;i++){
        j = index[i];
        data1[j] = data2[i] * data3[i];
    }
}
```



Vectorisation example

- This loop doesn't vectorise either:

```
do j = 1,N
  x = xinit
  do i = 1,N
    x = x + vexpr(i,j)
    y(i) = y(i) + x
  end do
end do
```

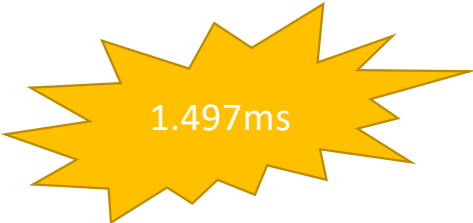
- Compiler will vectorise inner loop by default
 - Dependency on x between loop iterations

```
do j = 1,N
  x(j) = xinit
end do
do j = 1,N
  do i = 1,N
    x(j) = x(j) + vexpr(i,j)
    y(i) = y(i) + x(j)
  end do
end do
```



Example

```
16. + 1-----<   do j = 1,N
17.   1             x = xinit
18. + 1 r4-----<   do i = 1,N
19.   1 r4           x = x + vexpr(i,j)
20.   1 r4           y(i) = y(i) + x
21.   1 r4----->   end do
22.   1----->   end do
```



1.497ms

ftn-6254 ftn: VECTOR File = bufpack.F90, Line = 16

A loop starting at line 16 was **not vectorized** because a recurrence was found on "y" at line 20.

ftn-6005 ftn: SCALAR File = bufpack.F90, Line = 18

A loop starting at line 18 was **unrolled 4 times**.

ftn-6254 ftn: VECTOR File = bufpack.F90, Line = 18

A loop starting at line 18 was not vectorized because a recurrence was found on "x" at line 19.

```

38.  Vf-----<  do j = 1,N
39.  Vf          x(j) = xinit
40.  Vf----->  end do
41.
42.  ir4-----<  do j = 1,N
43.  ir4 if--<    do i = 1,N
44.  ir4 if      x(j) = x(j) + vexpr(i,j)
45.  ir4 if      y(i) = y(i) + x(j)
46.  ir4 if-->    end do
47.  ir4----->  end do

```

x promoted to vector:
trade slightly more memory
for better performance

ftn-6007 ftn: SCALAR File = bufpack.F90, Line = 42

A loop starting at line 42 was **interchanged** with the loop starting at line 43.

ftn-6004 ftn: SCALAR File = bufpack.F90, Line = 43

A loop starting at line 43 was **fused** with the loop starting at line 38.

ftn-6204 ftn: VECTOR File = bufpack.F90, Line = 38

A loop starting at line 38 was **vectorized**.

ftn-6208 ftn: VECTOR File = bufpack.F90, Line = 42

A loop starting at line 42 was **vectorized** as part of the loop starting at line 38.

ftn-6005 ftn: SCALAR File = bufpack.F90, Line = 42

A loop starting at line 42 was **unrolled 4 times**.

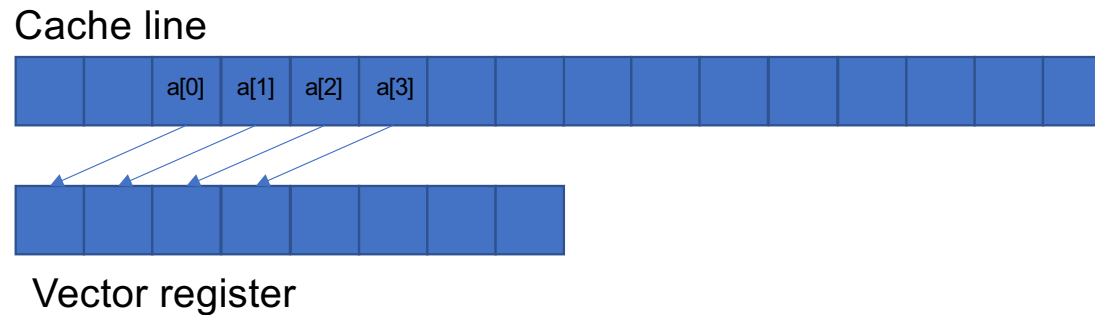
1.089ms

-37%



Data alignment

- When vectorising data aligned data is essential for performance



- Unaligned data
 - May require multiple data loads, multiple cache lines, multiple instructions
 - Will generate 3 different versions of a loop: peel, kernel, remainder
- Aligned data
 - Minimum number of data loads/cache lines/instructions
 - Will generate 2 different versions of a loop: kernel and remainder

Aligned data

- Aligned data is best
 - e.g. AVX vector loads/stores operate most effectively on 32-bytes aligned address
 - need to either let the compiler align the data....
 - ..or tell it what the alignment is
- Unit stride through memory is best

Align data

- Align on allocate/create (dynamic)

- `_mm_malloc, _mm_free`
- `float *a = _mm_malloc(1024*sizeof(float), 64);`
- align attribute (at definition, not allocation)
- `real, allocatable :: A(1024)`
- `!dir$ attributes align : 64 :: a`

- Align on definition (static)

```
float a[1024] __attribute__((aligned(64)));
real :: A(1024)
!dir$ attributes align : 64 :: a
```

- Common blocks in Fortran

- It's not possible to use directives to align data inside a common block
- Can align the start of a common block

```
!DIR$ ATTRIBUTES ALIGN : 64 :: /common_name/
```

- Up to you to pad elements inside common block

- Derived types

- May need to use `SEQUENCE` keyword and manually pad to get correct alignment

Multi-dimensional alignment



- Need to be careful with multi-dimensional arrays and alignment
 - If you `_mm_malloc` each dimension then it should be fine
 - If you do a single dimension `_mm_malloc` there may be issues:

```
float* a = _mm_malloc(16*15(sizeof(float), 64);  
for(i=0;i<16;i++){  
#pragma clang loop vectorize(enable)  
    for(j=0;j<15;j++){  
        a[i*15+j]++;  
    }  
}
```



Inform on alignment

- For non-static data, as well as aligning data, need to tell compiler it is aligned
- Number of different ways to do this
- Alignment of data inside a loop
 - Specify all data in the loop is aligned

```
#pragma vector aligned
!dir$ vector aligned
```

- Alignment of an array
 - Specify, for code after the alignment statement, a specific array is aligned

```
__assume_aligned(a, 64);
!dir$ assume_aligned a: 64
```

- May also need to define to properties of loop scalars

```
__assume(n1%16==0);
for(i=0;i<n;i++){
    x[i] = a[i] + a[i-n1] + a[i+n1];
}
!dir$ assume(mod(n1,16).eq.0)
```

- Also can use OpenMP simd clause
 - Specify array is aligned for simd loop

```
#pragma omp simd aligned(a:64)
!omp$ simd aligned(a:64)
```

Fortran data

- Different ways of passing data to subroutines can affect performance
- Explicit arrays

```
subroutine vec_add_mult(A, B, C)
real, intent(inout), dimension(1024) :: A
real, intent(in), dimension(1024) :: B, C
```

- Compiler generates subroutine code based on contiguous data
 - Packing/unpacking required to do this is done by the compiler at caller level
 - May be overhead associated with this
- Need to tell the compiler the arrays are aligned (i.e. `!dir$ assume_aligned` or `!dir$ vector aligned`)
- Same for arrays where array size is passed as an argument to the routine

Fortran data

- Assumed size arrays

```
subroutine vec_add_mult(A, B, C)
real, intent(inout), dimension(:) :: A
real, intent(in), dimension(:) :: B, C
```

- Compiler will generate different versions of the code, with and without contiguous functionality
 - Different versions may show up in the vector reports from the compiler
 - If there are too many different potential versions not all of them will necessarily be generated
 - The fall back version (none unit stride, not vectorised) will be used in this case for inputs that don't match any of the other versions
- Choice which is used made at runtime
- Still need to tell the compiler the arrays are aligned

Fortran data

- Assumed shape arrays

```
subroutine vec_add_mult(A, B, C)
real, intent(inout), dimension(*) :: A
real, intent(in), dimension(*) :: B, C
```

- Compiler generates subroutine code based on contiguous data
 - Packing/unpacking required to do this is done by the compiler at caller level
 - May be overhead associated with this
- Still need to tell the compiler the arrays are aligned

Fortran Indirect addressing

- Indirect addressing code can have some strange affects on vectorisation

```
subroutine vec_add_mult(A, B, C, index)
real, intent(inout), dimension(1024) :: A
real, intent(in), dimension(1024) :: B, C
integer, intent(in), dimension(1024) :: index
integer :: I
```

- Following has flow dependency (needs `ivdep` directive)

```
do i=1,n
  a(index(i)) = a(index(i)) + b(index(i)) * c(index(i))
end do
```

- Uses gather and scatter operations to pack/unpack indexed locations
- Following creates array temporary for right hand side evaluation

```
a(index(:)) = a(index(:)) + b(index(:)) * c(index(:))
```

- Ends up creating 2 loops

```
temp(:) = a(index(:)) + b(index(:)) * c(index(:))
a(index(:)) = temp(:)
```

- Uses gather/scatter in both loops

OpenMP 4.0 SIMD directives

- Many compilers support their own sets of directives to assist the compiler to vectorise loops.
 - useful but not portable
- OpenMP 4.0 contains a standardised set of directives

Portable SIMD directives

- Use **simd** directive to indicate a loop should be vectorised

```
#pragma omp simd [clauses]
```

or

```
!$omp simd [clauses]
```

- Executes iterations of following loop in SIMD chunks
- Loop is *not* divided across threads
- SIMD chunk is set of iterations executed concurrently by SIMD lanes
- Not a hint! Programmer is asserting independence of iterations.

- Clauses control data environment, how loop is partitioned
- **safelen(length)** limits the number of iterations in a SIMD chunk.
- **linear** lists variables with a linear relationship to the iteration space (induction variables)
- **aligned** specifies byte alignments of a list of variables
- **private, lastprivate, reduction** and **collapse** have usual meanings.
- Also **declare simd** directive to generate SIMDised versions of functions.
- Can be combined with loop constructs (parallelise and vectorise)
 - **#pragma omp for simd**

Example 1

```
do il=ilg,nm,-1
  jl = il - nm + 1
  if (il-nm.gt.0) then
    d1 = dsqrt(dble((il-nm)*(il+nm)))
    d2 = dble(il)
    ddlf(jl) = (d2*dc*dlf(jl) - d1*dlf(jl-1)) / ds
  else
    ddlf(1) = nm * dc * dlf(1)/ds
  endif
enddo
```

if test prevents vectorisation

Example 1

```
do il=ilg,nm+1,-1
  jl = il - nm + 1
  d1 = dsqrt(dble((il-nm)*(il+nm)))
  d2 = dble(il)
  ddlf(jl) = (d2*dc*dlf(jl) - d1*dlf(jl-1)) / ds
enddo
ddlf(1) = nm * dc * dlf(1)/ds
```

Vectorises!

Example 2

```
do il=2,ilg-nm
  d0 = il+2*nm
  d1 = dble((il-1) * (d0-1))
  d2 = dble(il * d0)
  dbeta = dsqrt(d1/d2)
  d1 = dble(2*(il+nm)-1)
  dalpha = d1/dsqrt(d2)
  dlf(il+1) = dalpha*dlf(il)*dc - dbeta*dlf(il-1)
enddo
```

dependency prevents vectorisation

Example 2

```
do il=2,ilg-nm
  d0 = il+2*nm
  d1 = dble((il-1) * (d0-1))
  d2 = dble(il * d0)
  dbeta(il) = dsqrt(d1/d2)
  d1 = dble(2*(il+nm)-1)
  dalpha(il) = d1/dsqrt(d2)
enddo
```

This loop now vectorizes!

Scalars expanded to arrays.
For large trip counts, loop
blocking would avoid extra
cache misses.

```
do il=2,ilg-nm
  dlf(il+1) = dalpha(il)*dlf(il)*dc - dbeta(il)*dlf(il-1)
enddo
```

Example 3

```
do k = 0, nlayers-1
  do df = 1, ndf2
    x_e(df) = x(map2(df)+k)
  end do
  ik = (cell-1)*nlayers + k + 1
  lhs_e = matmul(matrix(:, :, ik), x_e)
  do df = 1, ndf1
    lhs(map1(df)+k) = lhs(map1(df)+k) + lhs_e(df)
  end do
end do
```

Matrix dimensions ndf1 and ndf2 are small (e.g. 1,2 or 6) but nlayers is ~100. Inner loops can't utilize vector instructions.

Example 3

```
do df = 1, ndf1
  do df2 = 1, ndf2
    do k = 0, nlayers-1
      ik = (cell-1)*nlayers+k+1
      lhs(map1(df)+k) = lhs(map1(df)+k) + &
        matrix(ik, df, df2)*x(map2(df2)+k)
    end do
  end do
end do
```

Matrix array reordered so
memory accesses are
contiguous.

Long inner loop now
vectorizes well